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!pip install -q opencv-python matplotlib numpy
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import cv2
import numpy as np
import matplotlib.pyplot as plt

def show(title, img):
    plt.figure(figsize=(5,5))
    plt.imshow(img, cmap='gray')
    plt.title(title)
    plt.axis("off")
    plt.show()

def negative(img):
    return 255 - img

def log_transform(img):
    img = img.astype(np.float32)
    c = 255 / np.log(1 + np.max(img))
    return np.uint8(c * np.log(1 + img))

def gamma_correction(img, gamma):
    img_norm = img / 255.0
    corrected = np.power(img_norm, gamma)
    return np.uint8(corrected * 255)

def contrast_stretch(img):
    min_val = np.min(img)
    max_val = np.max(img)
    stretched = (img - min_val) * (255 / (max_val - min_val + 1e-10))
    return stretched.astype(np.uint8)

def hist_equalize(img):
    return cv2.equalizeHist(img)
```

```
image_paths = [
    "/content/Picture1.png",
    "/content/Picture2.png",
    "/content/Picture3.png",
    "/content/Picture4.png",
    "/content/Picture5.png"
]

print("Processing all 5 images...\n")

for path in image_paths:
    print("IMAGE:", path)

    img = cv2.imread(path, 0)
    if img is None:
        print("ERROR: File not found:", path)
        continue

    show("Original Image", img)

    img_neg = negative(img)
    show("Negative Transform", img_neg)

    img_log = log_transform(img)
    show("Log Transform", img_log)

    img_gamma1 = gamma_correction(img, 0.5)
    img_gamma2 = gamma_correction(img, 2.0)
    show("Gamma 0.5 (Brightening)", img_gamma1)
    show("Gamma 2.0 (Darkening)", img_gamma2)

    img_stretch = contrast_stretch(img)
    show("Contrast Stretching", img_stretch)

    img_hist = hist_equalize(img)
    show("Histogram Equalization", img_hist)

    print("\nDone for:", path, "\n\n")
```